

Visual Content-Based Music Recommendation

Group Project Report

Team Name: miracleAligners

Institution: Vietnamese-German University (VGU)

Course: Introduction to AI & Data Science

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1 Team Information

Table 1: Team Members and Responsibilities

Name	Student ID	Role / Core Responsibility
MAI HUYNH MINH THU	104240759	Main Researcher + Evaluation
NGUYEN HOANG GIA AN	104240384	G1 Main Frontend Dev
NGUYEN NGOC MINH VY	104240475	G2 FastAPI Setup + Misc. (Report)
TRAN GIA LINH	104240762	G2 Dataset
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2 Abstract

This project presents an interactive, visual content-based music recommendation framework that bridges digital image aesthetics with context-aware audio playlists. Instead of relying on historically tracked user profiles or manual text inputs, the platform deploys a pre-trained Contrastive Language-Image Pre-training (CLIP) framework to analyze uploaded images and classify them into seven target aesthetic categories (“vibes”). Concurrently, a structured music pool database was built by mapping audio tracks to these corresponding aesthetic styles.

To capture nuanced images where a single mood tag is insufficient, we engineered a relative Margin-Gap Optimization rule that dynamically flags multi-vibe visual states. The evaluation framework was benchmarked against a custom testing collection of 1,241 human-verified ground-truth images. The complete pipeline achieved a final classification accuracy of 63.58%, outperforming a baseline random guess probability of 14.28% by more than fourfold. The finalized web app prototype displays detected visual styles and serves real-time matching audio recommendations through an unauthenticated background media player interface.

3 Introduction

3.1 Problem Context

Social media platforms like TikTok and Instagram have revolutionized digital consumption by pairing user-generated images or short videos with contextually relevant audio tracks. Often, the selected track’s emotional tone, rhythm, and thematic markers align perfectly with the visual

elements. However, behind these commercial platforms are massive collaborative filtering models fueled by endless streams of user interaction data. For standalone applications or small-scale web ecosystems, this data abundance is unavailable.

Manually browsing millions of tracks to find a song that matches the exact "vibe" of a single photograph is an inefficient, tedious process. This problem limits the speed of content production and exhausts the creative energy of users. Solving this issue is critical because automated cross-modal synchronization drastically saves time, unlocks new artistic combinations, and elevates overall user satisfaction within digital platforms.

3.2 Project Goals & Target Audience

The direct beneficiaries of this system are independent content creators, digital marketers, and general social media users who require immediate, high-quality audio backdrops for their media. Additionally, normal users seem to be significantly benefiting from this feature, as a large number of people show excitement towards its appearance on several platforms. The primary goal of this project is to emulate this modern industrial capability at a smaller scale by building an AI system that translates visual imagery into highly accurate, mood-matched music recommendations.

4 Related Work / Literature Review

Traditional image-based music recommendation workflows rely on Convolutional Neural Networks (CNNs), such as ResNet or EfficientNet, to extract concrete objects or dominant color values from an image layout. While highly effective at objective vision tasks (like identifying a desk or a car), these traditional models cannot process abstract, subjective cultural concepts like "lofi," "cinematic," or "edgy" visual styles.

Our system approaches this problem differently by using a multi-modal foundation architecture. Rather than treating vision and text as completely separate domains, our implementation uses a shared vector space mapping structure. Furthermore, where existing commercial systems use restrictive API keys that require explicit user logins, our architecture implements an open fallback strategy to deliver continuous, unauthenticated audio streams directly through the web browser.

5 Problem Statement

Given a user-submitted image file stream, our system aims to predict the underlying aesthetic mood category and recommend a streamable, matching playlist.

- **Input Configuration:** A raw user-uploaded image file stream supporting .jpg, .jpeg, .png, or .webp extensions.
- **Expected Output Configuration:** A dynamic backend JSON payload containing detected single or multi-vibe labels, calculated prompt confidence weights, and 5 corresponding track objects complete with operational audio preview paths.
- **Core Task Selections:** Recommendation System, Computer Vision, and Natural Language Processing.

6 Dataset

Our system relies on a dual-source dataset structure consisting of an offline human-validated testing image collection and an indexed database of contextual music tracks.

6.1 Image Collection Specifications

Our custom validation data collection consists of $\sim 1,300$ raw internet images across 7 specific aesthetic categories: **melancholic/sad**, **hype/energetic**, **romantic/inlove**, **edgy/individualistic**, **cinematic/epic**, **chill/lofi**, and **soft/gentle**.

6.2 Music Database Specifications

The track database holds a balanced collection of roughly 200 to 300 tracks per aesthetic category, creating a stable, local music repository.

Table 2: Aesthetic-to-Audio Mapping Profile Matrix

Aesthetic Category	Mapped Spotify Genres / Playlists	Core Audio Characteristics
chill/lofi	Lo-Fi Beats, Chillhop, Study Beats	Low BPM (70–90), high acousticism, low energy, warm frequencies.
hype/energetic	EDM, Club Hits, Hip Hop, Gym/Workout	High BPM (120–140+), extreme danceability, high energy, heavy bass.
melancholic/sad	Sad Indie, Acoustic Folk, Dark Folk	Low-to-mid BPM, low valence (sadness scales), prominent acoustic instruments, muted tones.
edgy/individualistic	Modern Darkwave, Post-Punk, Heavy/Metal/Alternative Rock, Synth	Mid-to-high BPM, heavy synthetic basslines, dark/reverb-heavy tonal qualities.
cinematic/epic	Orchestral Score, Trailer Music, Soundtracks	Highly dynamic BPM, extreme volume variance, rich instrumental layers, sweeping arrangements.
romantic/inlove	New Age Piano, Pop Ballads, Soft R&B	Mid BPM, high emotional valence, gentle melodies, warm/intimate vocal tracking.
soft/gentle	Ambient, Minimalist Piano, Acoustic Pop	Very low-to-mid BPM, extremely low electronic energy, high peaceful/soft acoustic elements.

6.3 Programmatic Pivot 1

- **Original Design Proposal:** Scrape track lists directly out of curated public Spotify playlists using the playlist tracks endpoint (`v1/playlists/{playlist_id}/tracks`).
- **Discovered Reality:** Modern Spotify web API security protocols require complex user token authentication blocks for playlist track reads. Attempting to pull large batches of nested tracks programmatically without an active user log-in triggers strict **403 Forbidden** errors.
- **The Decision Change Solution:** We rewrote our dataset ingestion pipeline to use an unauthenticated global catalog search endpoint (`v1/search?type=track`). By transforming our 7 design aesthetic styles into explicit text keywords, the script queried individual track entries directly from the global database, bypassing playlist authentication limits and ensuring a balanced dataset.

7 Methodology

Our architecture links an interactive frontend UI with an active machine learning inference engine through an asynchronous API communication layer.

7.1 Multi-Modal Vision Framework

We implemented OpenAI’s `openai/clip-vit-base-patch32` foundation model. CLIP projects image pixels and text prompts into a shared high-dimensional vector space. The system feeds the uploaded image into a Vision Transformer (*Image Encoder*), while our 7 pre-defined labels are converted into descriptive phrase embeddings via a Text Transformer (*Text Encoder*). Suitability is determined by calculating the Cosine Similarity between the image vector ($\vec{I}$) and prompt vectors ($\vec{T}$):

$$\text{Similarity} = \frac{\vec{I} \cdot \vec{T}}{\|\vec{I}\| \|\vec{T}\|} \quad (1)$$

The model output provides 7 comparative raw confidence weights.

7.2 Programmatic Pivot 2:

- **Original Design Proposal:** Apply a classic fixed absolute cutoff threshold (such as requiring a label score to exceed 50%) to handle classification outputs.
- **Discovered Reality:** A standard Softmax layer scales prediction scores so they compete and always sum up to exactly 100%. If an uploaded image is a natural hybrid of two overlapping styles (e.g., scoring 46% `chill/lofi` and 44% `melancholic/sad`), it is mathematically impossible for either class to cross a 50% absolute threshold. This forces the engine to drop the secondary mood entirely.
- **The Decision Change Solution:** We replaced the absolute cutoff logic with a **Relative Margin-Gap Optimization Rule**. The server sorts all 7 scores, flags the top two categories, and evaluates the mathematical distance between them:

$$\text{Gap} = \text{Score}_{\text{primary}} - \text{Score}_{\text{secondary}} \quad (2)$$

We configured our threshold parameter to ≤ 0.15 (15%). If the computed gap is small, the system triggers a **Multi-Vibe Hybrid State**, automatically serving a blended playlist containing 3 tracks from the primary category and 2 tracks from the secondary pool. If the gap is wide, a single dominant mood takes over and serves all 5 tracks from the winning class.

7.3 Programmatic Pivot 3:

- **Original Design Proposal:** Stream audio files directly to the browser UI using the default `preview_url` property returned by the Spotify API metadata payload.
- **Discovered Reality:** Spotify completely deprecated its open track preview audio streams. The `preview_url` parameter across the API response now returns a `null` object, rendering the frontend media player entirely silent.
- **The Decision Change Solution:** We engineered an asynchronous JavaScript integration adapter inside `recommendationApi.js`. When the backend returns the recommended tracklist data, our UI intercept layer captures each song’s title and artist name strings, formats them into background query URLs, and checks them concurrently against Apple’s

open, unauthenticated **iTunes Search API**. The adapter extracts a working 30-second audio snippet link and attaches it to the track object on the fly, keeping the frontend streaming player active.

8 Experiments and Results

8.1 Evaluation Setup & Ground Truth Generation

To evaluate our system objectively without mixing ambiguous testing terms, we set up an offline validation loop. Three human judges (Vy, Trâm, Linh) independently reviewed our raw image database. To automate consensus tracking across subjective aesthetics, we implemented a multi-column majority-voting matrix formula:

```
=IF(COUNTA(B2:D2)<3,"Waiting...",IF(C2=D2,C2,IF(C2=E2,C2,IF(E2=D2,E2,"DISCARD_TIE"))))
```

If all three team members disagreed completely on a photo, the image was flagged as DISCARD.TIE and removed from the evaluation pool to preserve ground-truth integrity. This rigorous filtering process left **1,241 clean images** for our performance benchmarking test suite.

8.2 Final Performance Metrics Report

The evaluation script ran our CLIP inference model across every verified ground-truth image folder path, yielding the following performance profile:

- **Total Discarded Voting Ties (Skipped): 0**
- **Clean Images Evaluated: 1,241 Samples**
- **Correct AI Classifications: 789 Images**
- **Final System Testing Accuracy: 63.58%**
- **Comparative Baseline Performance (Random Guess Floor): 14.28%** (1 out of 7 choices)

8.3 Core Performance Breakdown (Confusion Matrix Interpretation)

The confusion matrix tracks true positive counts along with semantic crossovers:

Table 3: CLIP Model Aesthetic Confusion Matrix Across 7 Core Vibes

Actual Human Label	chill/lofi	cinematic	edgy	hype	melancholic	romantic	soft
chill/lofi	55	21	9	13	17	24	18
cinematic/epic	11	125	5	1	11	12	16
edgy/indie	10	19	87	61	4	4	0
hype/energetic	6	19	6	157	0	1	0
melancholic/sad	30	30	14	0	111	9	8
romantic/inlove	0	7	2	5	1	155	14
soft/gentle	8	11	12	6	3	4	99

The matrix exhibits a prominent diagonal line of high true positive counts, confirming that the model effectively maps abstract styles. Minor semantic bleeding occurs predominantly between the **edgy/individualistic** and **hype/energetic** categories (61 misclassifications), as well as between **melancholic/sad** and **chill/lofi** groupings. Because these styles share overlapping visual elements—such as low color saturation, high flash artifacts, or muted shadows—this behavior matches natural human artistic interpretation.

9 Discussion

9.1 What Worked Well

1. **Multimodal Architecture Efficiency:** Using a zero-shot model allowed our platform to interpret complex, social-media-style imagery instantly without requiring local GPU training clusters or manually labeling a massive training set.
2. **Resilient Error Control:** The relative Margin-Gap rule handles ambiguous inputs gracefully. Instead of making an incorrect single guess, flat prediction scores trigger a balanced, diverse hybrid playlist for the end user.

9.2 Limitations

1. In images with heavily desaturated tones or uniform lighting, the model's confidence vectors cluster too closely, which increases classification error rates across subjective boundaries.
2. The inference model is structurally bounded by the 7 pre-defined prompt strings configured in the python codebase.
3. A significant limitation of the current recommendation framework is the absence of track-level scoring granularity. Instead of computing an individualized compatibility score for each song based on localized acoustic features, the system applies a uniform top-down assignment mapping inherited directly from the vision classifier.

As a direct consequence, if an uploaded image is classified under the **edgy/individualistic** vibe with a confidence score of 50%, every single track fetched from that respective pool is uniformly presented to the user interface as a 50% match. This structural bottleneck prevents the system from ranking or differentiating the contextual relevance of individual songs within the same category cluster, neglecting the natural intra-class variances of musical features.

9.3 Future Modifications

1. We can add or adjust categories without retraining the model by simply updating the text prompt strings in our backend array.
2. Future updates can lock the core CLIP weights and train a lightweight classification layer specifically on our images to push classification accuracy past 75%.
3. To resolve the uniform confidence scoring limitation, the system can be upgraded from a flat categorical database lookup to a fine-grained, two-stage recommendation pipeline. This implementation introduces localized track sorting based on unique multi-modal feature alignments.

10 Output / Final Output

Our team successfully built and delivered the following core deliverables:

- **Technical Documentation Report:** Full system implementation and architectural review.
- **Source Code Repository:** Modular Python backend scripts alongside our frontend React framework layout.
- **Prediction Engine Pipeline:** Operational multi-vibe vision extraction framework.

- **Data Dashboard Visualizations:** Automated class distribution charts and confusion matrix plots.
- **Web Prototype Application:** An active, dark-mode web prototype featuring dynamic color palettes and an unauthenticated background streaming player.

11 Conclusion

This project successfully developed an alternative to rigid, keyword-driven music discovery engines. By using OpenAI’s zero-shot CLIP framework, we built a full-stack web application capable of mapping complex user imagery to tailored, multi-vibe audio playlists. Rigorous benchmarking against 1,241 human-verified images achieved a final system accuracy score of 63.58%, significantly outperforming the baseline random guess floor of 14.28%. The system handles complex real-world imagery gracefully, creating a stable, responsive multimedia streaming experience.

12 References

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